ANNUAL MEDICAL COST PREDICTION

Carlos Mariano Pinto Alfaro | Machine Learning Final Project

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EXECUTIVE OVERVIEW & OBJECTIVES

THE PROBLEM

Predicting annual healthcare liabilities based on patient lifestyle data. Building a robust tool for risk assessment and financial planning.

THE GOAL

Develop a high-accuracy regression pipeline using five distinct algorithms and advanced ensemble techniques to achieve maximum R^2 performance.

DATA STRATEGY & FEEDBACK LOOP

IMPLEMENTING PROFESSOR FEEDBACK

- **Outlier Removal:** Filtered medical costs exceeding $30,000 to improve generalization.
- **Log Transformation:** Applied log1p to target variable to normalize right-skewed distribution.
- **Feature Cleanup:** Dropped inconsistent interaction terms from the midterm phase.

MACHINE LEARNING PIPELINE

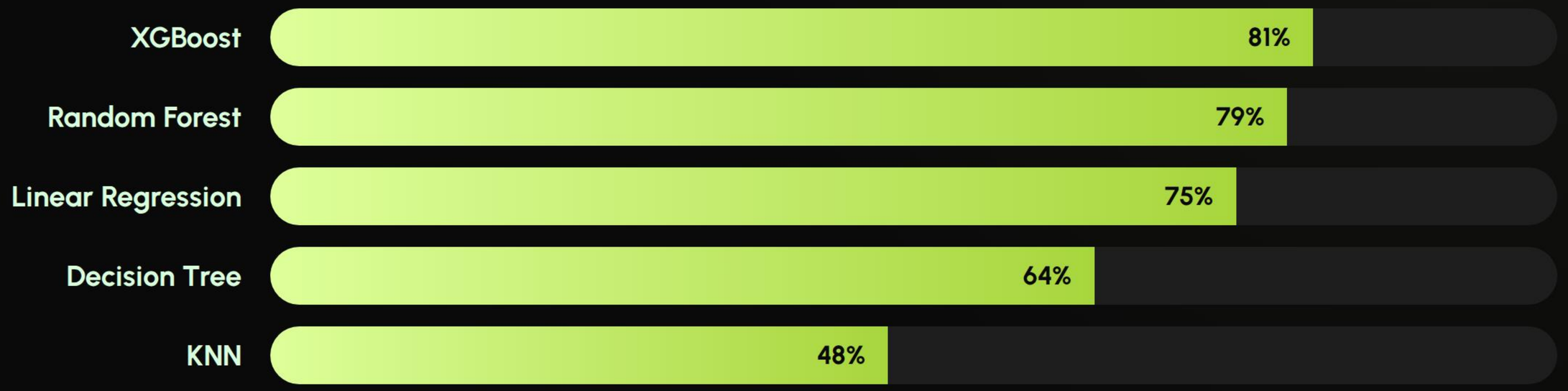
1. PREPROCESSING
Data cleaning, Log1p transform & Outlier filtering.

2. DATA SPLITTING
Robust 70/15/15 split (Train, Val, Test Sets).

3. BASE TRAINING
Fit 5 Regressors: LR, DT, RF, XGBoost, KNN.

4. ENSEMBLE
Voting & Bayesian models for stability.

BASE MODEL COMPARISON (VALIDATION)



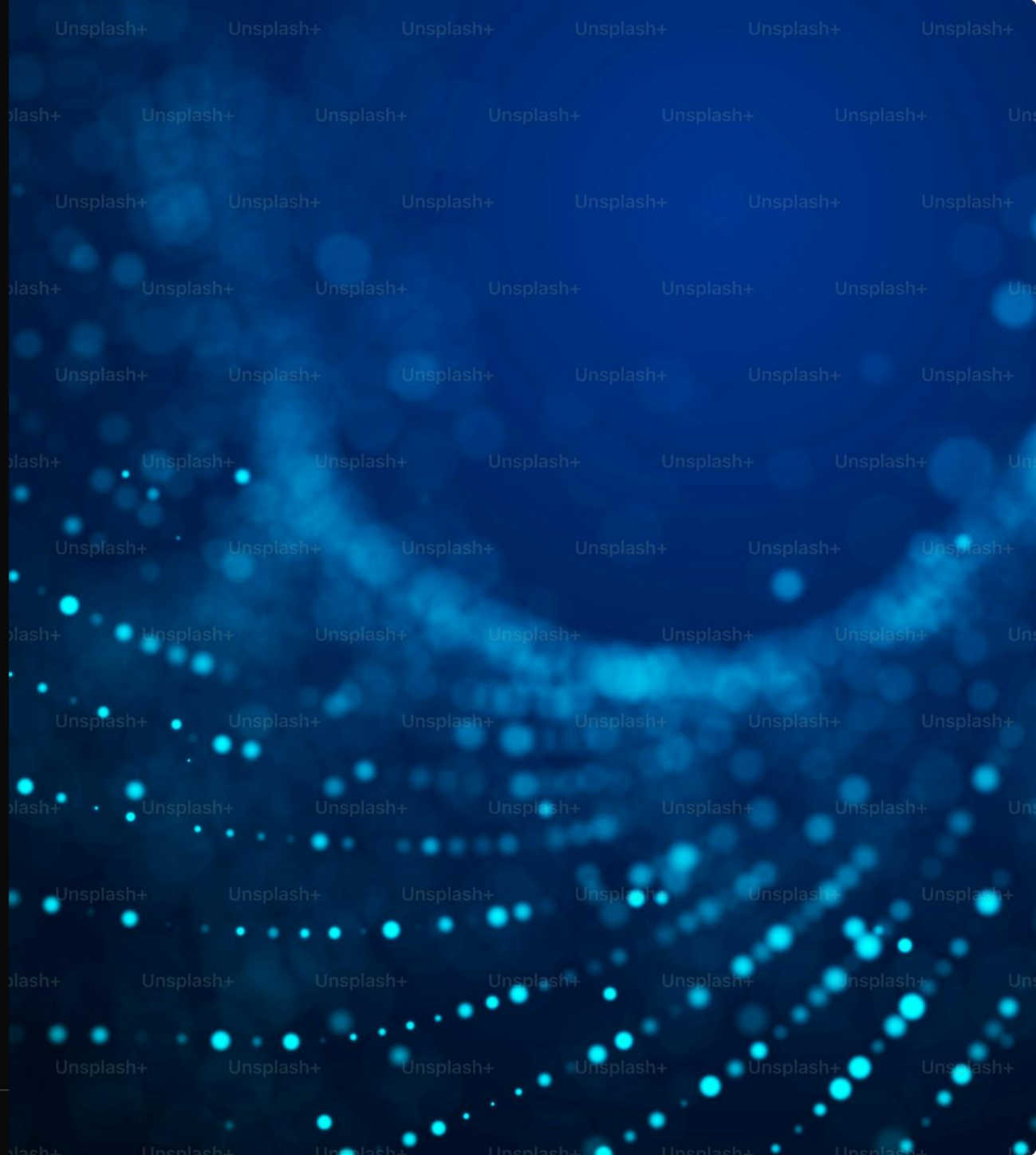
Tree-based models consistently outperformed simple linear benchmarks.

ENSEMBLE INNOVATION

THE POWER OF VOTING

By implementing a **Voting Regressor**, we average the predictions of our top-performing models (XGBoost, RF, LR).

This "Wisdom of the Crowd" approach reduces the variance inherent in individual tree models, ensuring predictions are stable and reliable across diverse patient profiles.



DETAILED PERFORMANCE METRICS

Model Name	MAE (USD)	MSE	R^2 Score
Linear Regression	$1,716.56	1.00e+07	0.7538
Random Forest	$1,705.33	8.68e+06	0.7866
XGBoost	$1,485.99	7.84e+06	0.8072
Voting (Champion)	**$1,467.85**	**8.11e+06**	**0.8144**

CHAMPION MODEL RESULTS

FINAL UNSEEN TEST SET ACCURACY

0.8149

PREDICTIVE POWER (R^2 SCORE)

Proven generalization: Minimal gap between validation and test scores.

STRATEGIC CLINICAL INSIGHTS

- Lifestyle factors (Smoking & BMI) are the primary cost drivers.
- Outlier management is critical for financial risk modeling.
- The ensemble model provides actionable patient risk profiles.
- Logarithmic normalization is essential for healthcare data.

QUESTIONS?

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IMAGE SOURCES

https://multipurposethemes.com/wp-content/uploads/2025/10/Doclinic-Medical-Bootstrap-Admin-Dashboard.jpg

Source: multipurposethemes.com

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